MINI ASSESSMENT - LAB REPORT

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Date: 11-09-2026

1. AIM

To create and implement a simple rule-based conversational chatbot in Python that accepts user messages, identifies basic intents through conditional statements, and produces suitable responses during an interactive session.

2. PROCEDURE

1. Start a continuous `while True` loop so that the chatbot can interact with the user repeatedly.

2. Read each message from the keyboard using Python's `input()` function.

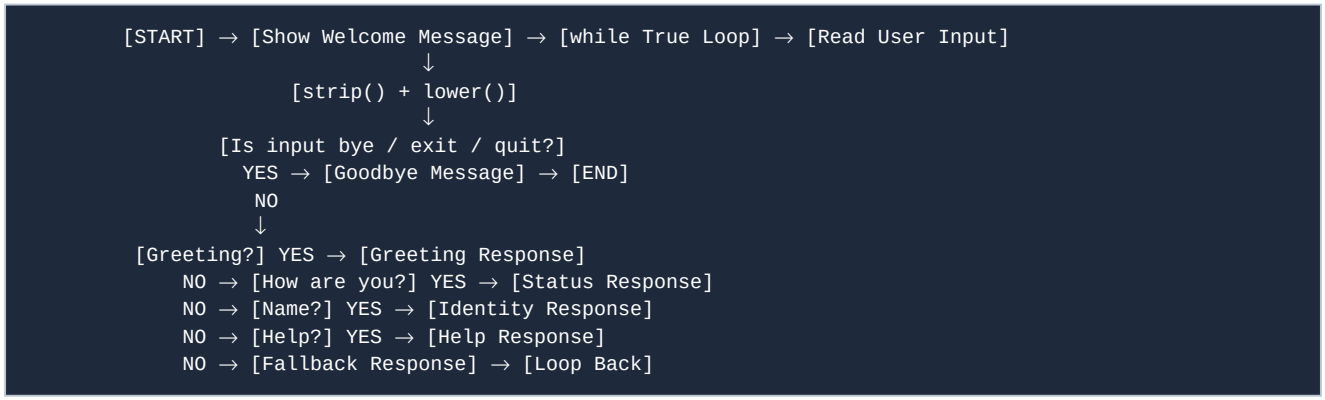
3. Clean the entered message using `strip()` and `lower()` so that extra spaces and capitalization do not affect matching.

4. Check whether the user entered a termination word such as `bye`, `exit`, or `quit`.

5. Use an `if-elif-else` decision structure to recognize greetings, status questions, identity questions, and help requests.

6. Use the final `else` branch to provide a friendly response when the input does not match any supported rule.

3. FLOWCHART



4. PYTHON CODE

```
def rule_based_bot():
    print("Bot: Hello! I am your Python assistant.")
    print("Type 'bye' or 'exit' whenever you want to stop.\n")

    while True:
        user_input = input("You: ").strip().lower()

        if user_input in ["bye", "exit", "quit"]:
            print("Bot: Goodbye! Have a great day!")
            break

        elif user_input in ["hello", "hi", "hey"]:
            print("Bot: Hello! How can I assist you?")

        elif "how are you" in user_input:
            print("Bot: I am doing well. Thanks for asking!")

        elif "your name" in user_input:
            print("Bot: I am a simple Python rule-based chatbot.")

        elif "help" in user_input:
            print("Bot: You can greet me, ask my status or name, "
                  "request help, or type 'bye' to exit.")

        else:
            print("Bot: Sorry, I do not recognize that request. "
                  "Try typing 'help'.")

if __name__ == "__main__":
    rule_based_bot()
```

4. PYTHON CODE (CONTINUED)

```
            # The decision branches above are evaluated in order.
            # The first matching rule produces the response.

            # Unrecognized input is handled by the final else block.
            # The loop continues until a termination word is entered.
```

5. RESULT

```
Bot: Hello! I am your Python assistant. Type 'bye' or 'exit' whenever you want to stop.
You: hi
Bot: Hello! How can I assist you?
You: how are you?
Bot: I am doing well. Thanks for asking!
You: what is your name?
Bot: I am a simple Python rule-based chatbot.
You: can you sing?
Bot: Sorry, I do not recognize that request. Try typing 'help'.
You: bye
Bot: Goodbye! Have a great day!
```

6. CONCLUSION

> The rule-based chatbot was successfully developed using Python conditional branching and a continuous while loop. It accepts user input, performs basic keyword-based intent matching, provides predefined responses, and terminates cleanly when an exit command is entered.